Adapting AI Foundation Models for Sub-seasonal Forecasting: Fine-Tuning Aurora for the Madden-Julian Oscillation

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***Abstract*—The Madden-Julian Oscillation (MJO) is the primary source of sub-seasonal predictability in the tropics, yet its complex dynamics and interactions with other atmospheric phenomena make it challenging for both General Circulation Models (GCMs) and purely statistical Deep Learning (DL) emulators to predict. We test an alternative: adapting a foundation model to simulate the MJO's evolution. We use Microsoft Aurora, a 1.3B parameter foundation model pre-trained on over one million hours of diverse atmospheric data, and fine-tune it to simulate the nonlinear dynamics of the MJO. Our method required adding MJO-critical variables, viz., Outgoing Longwave Radiation (OLR) and Total Column Water Vapor (TCWV) to Aurora's inputs, as they are essential for capturing the convective and moisture dynamics of the MJO. We developed a pipeline that injects these variables into the embedding space without catastrophic forgetting. A spectral loss function and OneCycleLR schedule helped the model adapt to these new variable distributions. Preliminary results on a micro-scale high-resolution dataset are promising, demonstrating that the fine-tuned model generates physically realistic, sharp convective structures and maintains stability over 10-day autoregressive rollouts. This initial success indicates potential for scaling to larger datasets using Low-Rank Adaptation (LoRA), which we hypothesize could enhance the reliability of long-range sub-seasonal forecasts, pending further empirical validation.**

***Index Terms*—Foundation Models, Madden-Julian Oscillation, Deep Learning, Sub-seasonal Forecasting, Transfer Learning, Atmospheric Physics.**

I. Introduction

The Madden-Julian Oscillation (MJO) is the primary mode of variability on the intra-seasonal scale in the tropical atmosphere, impacting regions from 30°N to 30°S. It is broadly characterized by a planetary-scale eastward propagation (4–8 $\mathrm{m\,s^{-1}}$) of deep convection and circulation anomalies with a periodicity of 30–60 days [1], [2]. To track this complex phenomenon, the atmospheric science community relies on the Real-time Multivariate MJO (RMM) index, developed by Wheeler and Hendon [3]. This index projects anomalies of Outgoing Longwave Radiation (OLR)[1] and zonal winds onto the two leading empirical orthogonal functions (EOFs) of tropical variability, yielding a two-dimensional phase space (RMM1 and RMM2) that describes the MJO's amplitude and longitudinal location.

As a primary source of predictability on the sub-seasonal timescale (around 2–8 weeks), the MJO is a critical bridge between medium-range weather forecasting and seasonal climate prediction. It is primarily active in the tropics, but its influence extends outward via global teleconnections, thereby modulating the frequency and intensity of various phenomena in the mid-latitudes such as atmospheric rivers, tropical cyclones, and extreme precipitation events [4]. Because of this, predicting the MJO is key to extended-range forecasting of potentially high-impact weather events.

In spite of its importance, the MJO remains difficult to forecast using traditional Numerical Weather Prediction (NWP) systems. General Circulation Models (GCMs) often struggle to sustain forecast skill for the MJO

[1]Note that we use *Top Thermal Radiation* (TTR) as a proxy for OLR; see §II-C.

beyond 10–14 days, limited by the rapid growth of initial condition errors and the imperfect parameterizations of moist convection [4]. This predictability gap has led researches to explore data-driven Deep Learning (DL) approaches. Prior work has largely treated MJO forecasting as a direct regression or *emulation* problem [5], including previous work by the Lu Group at Yale [6]. Within such frameworks, models are trained to map initial atmospheric anomalies at time t_0 directly to a target state or index at a future lead time $t_0 + \tau$ (e.g., τ =30 days) [7]. Model architecture varies but due to the nature of the image-like data, Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) are popular. Although these methods have indeed shown potential in capturing statistical correlations, they often lack physical consistency and require massive datasets to learn complex fluid dynamics from scratch. Moreover, recent work has highlighted the pitfalls of incorrect metric calculation in these statistical approaches, underlining the need for rigorous validation (cf. the Silini paper [8] and its response [9]).

In this work, we approach MJO forecasting by adapting a foundation model to predict the physical evolution of the atmosphere over time, contrasting with statistical emulation methods that rely on historical data correlations without simulating the underlying dynamics. We use **Microsoft Aurora**, a 1.3B parameter Foundation Model based on the 3D Swin Transformer architecture [10]. Because Aurora was trained on over one million hours of diverse climatological data, it implicitly encodes the governing equations for atmospheric flow, thermodynamics, and mass conservation. As opposed to training a model to "jump" 30 days into the future, we treat the MJO as an initial value physical problem, fine-tuning Aurora to simulate the evolution of the tropical atmosphere in discrete, physics-consistent time steps.

Our main contribution in this first stage has been the development of a data pipeline to adapt Aurora to the specific requirements of MJO forecasting. We identify a critical data mismatch in existing MJO benchmarks—viz., the reliance on daily anomaly fields—and demonstrate that Aurora requires high-resolution, absolute physical state variables to leverage its pre-training. We extend the Aurora embedding space to include key MJO variables, specifically Outgoing Longwave Radiation (OLR) and Total Column Water Vapor (TCWV), which serve as proxies for tropical convection and moisture mode dynamics, respectively. By fine-tuning the model on these raw physical fields, we show that the model can propoagate tropical convective structures in a stable and physically consistent way. The initial results we obtained from a micro-scale fine-tuning experiment show convergence and the ability to capture the spatial structure of MJO convection, establishing a robust pipeline for scaling to multi-decadal datasets.

As the project continues to evolve, we propose a strategic roadmap to evolve this pipeline from short-term adaptation to robust sub-seasonal forecasting. Our immediate next step involves scaling the ERA5 training dataset beyond one month to the decade scale. This will enable the model to capture inter-annual variability and ENSO teleconnections. To achieve stability over 30-day forecast horizons, we will implement fine-tuning via Low-Rank Adaptation (LoRA) [11] as recommended by the Aurora researchers [10]. By freezing the pre-trained backbone and training only low-rank matrices injected into the attention layers, we can computationally afford to train on long autoregressive rollouts (multi-step predictions) rather than just single time steps. Furthermore, leveraging the Lu Group's expertise in scientific machine learning (SciML), we intend to explore physics-informed constraints. This may involve augmenting the loss function with conservation penalties or developing a lightweight residual corrector network trained to minimize physical residuals (e.g., continuity equations) in the Aurora output, ensuring that long-range forecasts remain physically plausible.

II. Problem Formulation

Traditional data-driven approaches to MJO forecasting often frame the task as a direct regression problem wherein a model learns a mapping from an initial atmospheric state at time t_0 to a target index (RMM) at a future lead time $t_0 + \tau$. The primary advantage of this emulation approach is computational expediency; a model can be trained to simply "jump" over the chaotic intermediate dynamics. However, these methods bypass the underlying physics and rely on the model to infer long-term causal relationships from statistical correlations alone. As such, they are often data-inefficient, requiring decades of observational data to learn fundamental fluid dynamics from scratch.

In this work, we pivot from this prediction mechanism and formulate MJO forecasting as a prognostic, physics-based initial value problem. Our hypothesis is that a pre-trained model's learned physics can help simulate the MJO with higher fidelity, using less training data.

A. The MJO as an Initial Value Problem

We represent the state of the global atmosphere at a discrete time t by a high-dimensional state vector $\mathbf{X}(t)$. This vector encompasses a set of physical variables defined on a three-dimensional grid of latitude, longitude, and pressure levels. The evolution of the atmosphere is governed by a set of complex nonlinear partial differential equations (PDEs), specifically the Navier-Stokes equations on a rotating sphere coupled with thermodynamic state equations. A perfect prognostic model, $\mathcal{F}$, would describe this evolution such that:

$$\mathbf{X}(t + \Delta t) = \mathcal{F}(\mathbf{X}(t)), \tag{1}$$

where Δt is a discrete time step (in our case, 6 hours). Our goal is to leverage a pre-trained AI model, $\mathcal{M}_\theta$, to approximate this prognostic operator $\mathcal{F}$. By fine-tuning the model's parameters θ, we aim to produce a sequence of forecasts autoregressively:

$$\hat{\mathbf{X}}(t)(t_0 + \Delta t) = \mathcal{M}_\theta(\mathbf{X}(t_0) \tag{2}$$

$$\hat{\mathbf{X}}(t)(t_0 + 2\Delta t) = \mathcal{M}_\theta(\mathbf{X}(t_0 + \Delta t) \tag{3}$$

$$\vdots$$

$$\hat{\mathbf{X}}(t)(t_0 + N\Delta t) = \mathcal{M}_\theta(\mathbf{X}(t_0 + (N-1)\Delta t). \tag{4}$$

This approach ensures that each step in the forecast is physically consistent and allows the model to leverage its learned understanding of atmospheric dynamics, rather than attempting a statistically-driven "jump" across a long time horizon.

B. The Aurora Foundation Model

The feasibility of this approach rests on the choice of $\mathcal{M}_\theta$. We utilize Microsoft Aurora, a 1.3B parameter foundation model designed for high-resolution Earth system forecasting [10].

Aurora is distinct from standard task-specific models (e.g., CNNs or vanilla ViTs) in its training curriculum. It was pre-trained on a massive corpus of diverse data, including ERA5 reanalysis, CMIP6 climate simulations, and IFS operational forecasts. This exposure allows the model to learn a general-purpose representation of atmospheric dynamics, implicitly encoding the laws of conservation of mass, momentum, and energy, before it is ever exposed to our specific MJO task.

Architecturally, Aurora employs a 3D Swin Transformer backbone. This hierarchical ViT processes the atmosphere as a 3D volume, using shifted windows to compute self-attention. This architecture is particularly well-suited to our problem formulation because it explicitly models vertical interactions in the atmosphere, e.g., between surface heating and upper-level winds, which are critical for the convective processes that drive the MJO. However, utilizing a pre-trained model imposes strict constraints on the input state space $\mathbf{X}(t)$; Aurora expects certain variables and distributions, so we had to carefully define our state vector.

C. State Space Definition

To capture the thermodynamic and dynamic structures essential for MJO propagation, we define the atmospheric state vector $\mathbf{X}(t)$ using variables from the ERA5 reanalysis dataset. The input space is divided into three categories:

1) **Surface-level variables** (2D)
2) **Atmospheric variables** (3D, defined at multiple pressure levels)
3) **Static variables** (time invariant).

The standard Aurora backbone is pre-trained on a core set of meteorological variables including wind vectors, temperature, and geopotential height. However, standard meteorological variables alone are insufficient for robust MJO detection, as the phenomenon is defined largely by its convective signal and moisture envelope [3], [5]. Therefore, we extend the state space to include two critical surface-level proxies:

- **Top Thermal Radiation (TTR):** In the ERA5 dataset, TTR corresponds to *Outgoing Longwave Radiation* (OLR). Negative anomalies in OLR are the standard proxy for deep tropical convection (cloud tops are colder and emit less radiation) and are a required component for calculating the RMM index [3].
- **Total Column Water Vapor (TCWV):** The MJO is often described as a "moisture mode" disturbance, where instability is driven by moisture-convection feedback mechanisms [4]. TCWV integrates specific humidity across the atmospheric column, serving as a critical tracer for the eastward propagation of the moisture envelope.

Table I details the complete composition of the state vector $\mathbf{X}(t)$ used in our fine-tuning pipeline.

D. Goal: Domain Adaptation vs. Emulation

Unlike previous approaches that train models to predict anomalies—deviations from the mean climate—we task the model with predicting the *absolute physical state* of the atmosphere.

The injected variables, TTR and TCWV, possess statistical disitruvtions distinct from Aurora's pre-training set. For example, Geopotential (z) typically takes values

TABLE I: Definition of State Variables in the Fine-Tuned Model

Category	Code Name	Symbol	Units	Description & Role
Surface Variables	`2t`	T_{2m}	K	Air temperature at 2 meters. Captures surface heating and diurnal cycles.
	`10u`	u_{10}	m s^{-1}	Zonal (East-West) wind component at 10 meters. Critical for surface convergence and westerly wind bursts.
	`10v`	v_{10}	m s^{-1}	Meridional (North-South) wind component at 10 meters.
	`msl`	P_{msl}	Pa	Mean Sea Level Pressure. Captures large-scale circulation patterns.
	`ttr`	OLR	W m^{-2}	**[Extension]** Top Thermal Radiation. Proxy for deep convection and cloud cover.
	`tcwv`	$TCWV$	kg m^{-2}	**[Extension]** Total Column Water Vapor. Tracks the moisture envelope of the MJO.
Atmospheric Variables	`z`	Z	m^2 s^{-2}	Geopotential. Represents the gravitational potential energy; utilized to identify pressure systems.
	`u`	u	m s^{-1}	Zonal wind speed at pressure level p.
	`v`	v	m s^{-1}	Meridional wind speed at pressure level p.
	`t`	T	K	Air temperature at pressure level p.
	`q`	q	kg kg^{-1}	Specific Humidity. Mass of water vapor per unit mass of moist air.
Static Variables	`lsm`	-	0-1	Land-Sea Mask. Binary mask distinguishing ocean from land.
	`z_static`	Z_s	m^2 s^{-2}	Surface Geopotential (Orography). Represents terrain elevation.
	`slt`	-	Int	Soil Type. Categorical variable affecting surface friction and moisture.

Note: Atmospheric variables are provided at 13 vertical pressure levels: 50, 100, 150, 200, 250, 300, 400, 500, 600, 700, 850, 925, and 1000 hPa.

in the range of 10^4–10^5, while Specific Humidity (q) operates in the range of 10^{-2}–10^{-5}. TTR introduces negative flux values (hundreds of Watts per square meter).

Therefore, the core problem formulation is not just one of time-series prediction but also of domain adaptation. We must adapt the pre-trained embedding space of Aurora to accommodate these new physical variables without suffering from "catastrophic forgetting," where the model loses its pre-learned understanding of atmospheric physics. Our objective is to fine-tune the model $\mathcal{M}_\theta$ such that it can accurately propagate these new variables ($\hat{\mathbf{X}}_{\text{ext}}$) alongside the standard variables ($\hat{\mathbf{X}}_{\text{std}}$) over time:

$$\min_\theta \sum_t \left\| \hat{\mathbf{X}}_{ext}(t) - \mathbf{X}_{ext}(t) \right\| + \left\| \hat{\mathbf{X}}_{std}(t) - \mathbf{X}_{std}(t) \right\|. \tag{5}$$

III. Methodology

Adapting a pre-trained foundation model for a specific downstream task requires careful engineering to prevent the above-mentioned "catastrophic forgetting" of the pre-trained physics while allowing the model to adapt to new variables. Unlike training a small CNN from scratch, fine-tuning a 1.3B parameter model on high-resolution global data presents significant computational challenges. Our methodology consists of three distinct components: a robust data ingestion pipeline, a variable embedding extension strategy, and a physics-constrained hybrid loss formulation.

A. Data Engineering and Normalization

We utilize the ERA5 reanalysis dataset, which provides hourly estimates of atmospheric variables on a

0.25° global grid (721×1440 latitude-longitude points). For this study, we subsample the data to a 6-hourly temporal resolution to match the standard operational forecast cycle.

One challenge we faced was the distribution shift between pre-training and fine-tuning data. Aurora expects inputs to be strictly normalized. However, calculating global mean and variance statistics for TB-scale datasets is computationally prohibitive during runtime. We engineered a lazy-loading pipeline using `xarray` and `dask` that streams data directly from the disk during training. To handle normalization, we pre-computed static statistics (μ_v, σ_v) for each new variable v using a representative subset of the training period.

Special attention was required for the TTR variable. In the ERA5 dataset, TTR is provided as an accumulated energy flux (units Jm^{-2}) over the archiving interval. To get a physical *power* flux (units Wm^{-2}) consistent with the OLR variable used in the RMM index calculation, we use this transformation:

$$\text{TTR}_{Wm^{-2}} = \frac{\text{TTR}_{Jm^{-2}}}{\Delta t_{\text{sec}}}, \tag{6}$$

where $\Delta t_{\text{sec}} = 3600$. This ensures the numerical range of the variable aligns with the model's internal activation dynamics.

B. Variable Embedding Injection

The Aurora architecture utilizes a 3D Swin Transformer backbone [10]. The first layer of this net is a patch embedding layer that projects raw physical values into a high-dimensional latent space of dimension D.

Since TTR and TCWV were not present in the original pre-training corpus, the pre-trained patch embedding layer $\mathcal{E}_{\text{pre}}$ does not have weights associated with these channels. We introduce a set of new, randomly initialized embedding weights W_{new} for these specific variables. The input state $\mathbf{X}$ is split into the standard variables $\mathbf{X}_{\text{std}}$ and the new extension variables $\mathbf{X}_{\text{ext}}$. The projected latent representation $\mathbf{Z}$ is computed as:

$$\mathbf{Z} = \mathcal{E}_{\text{pre}}(\mathbf{X}_{\text{std}}) \oplus \text{Conv3D}(\mathbf{X}_{\text{ext}}; W_{\text{new}}), \tag{7}$$

where $\oplus$ denotes concatenation along the channel dimension. This way, the transformer can process the new MJO-critical variables with the standard ones without needing to change the core architecture.

C. Hybrid Loss Function

A common failure mode in data-driven weather forecasting is the production of "blurry" forecasts. Standard pixel-wise reconstruction losses, such as Mean Absolute Error (MAE, L_1) or Mean Squared Error (MSE), encourage the model to predict the conditional mean of the predictive distribution [12]. In the context of chaotic atmospheric systems like the MJO, where the exact location of a convective cell is uncertain, the mean prediction is often a smoothed, low-magnitude field that lacks physical realism.

To enforce structural sharpness and preserve the high-frequency features of tropical convection, we employ a hybrid loss function $\mathcal{L}_{\text{total}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{L_1} + \lambda \mathcal{L}_{\text{spec}}, \tag{8}$$

where $\mathcal{L}_{L_1}$ is the standard MAE weighted heavily in the tropical band (20°N–20°S) to focus the model on MJO dynamics. $\mathcal{L}_{\text{spec}}$ is a spectral loss term computed in the frequency domain using the 2D Fast Fourier Transform (FFT):

$$\mathcal{L}_{\text{spec}} = \frac{1}{N} \sum_k \left\| \text{FFT}(\hat{\mathbf{X}})_k - \text{FFT}(\mathbf{X})_k \right\|_1, \tag{9}$$

where $\text{FFT}(\cdot)$ denotes the 2D FFT operation and k indexes the frequency modes. This term penalizes discrepancies in the power spectrum, forcing the model to match the texture and spatial variance of the ground truth cloud, rather than just their position. We empirically found $\lambda = 0.05$ to provide a good balance between sharp outputs and stable training.

D. Optimization Strategy

Training was performed on a single NVIDIA A100 (80GB) GPU on the Yale Grace HPC. Due to the memory constraints of fine-tuning a 3D global model, we utilized two key memory optimization techniques:

1) **Mixed Precision Training (`bfloat16`):** We utilized `bfloat16` precision for the transformer backbone to reduce memory footprint and increase throughput, while keeping the encoder/decoder layers in `float32` to maintain numerical precision for the variable embeddings.
2) **Gradient Checkpointing:** We traded compute for memory by not storing intermediate activations during the forward pass, instead re-computing them during the backward pass.

Due to these constraints, the maximum physical batch size was limited to 1. In order to stabilize the noisy gradients associated with a batch size of 1, we implemented gradient accumulation with $N = 4$ steps, simulating an effective batch size of 4.

Finally, we observed that standard cosine annealing schedules led to loss spikes and training instability during fine-tuning. Instead, we implemented the `OneCycleLR` scheduler. This schedule rapidly increases the learning rate (LR) to a maximum of $\alpha_{\max} = 5 \times 10^{-5}$ to traverse the loss landscape's saddle points, before annealing to near-zero. This super-convergence behavior acts as a regularization technique, preventing overfitting on the relatively small validation dataset used in this proof-of-concept phase.

IV. Results & Discussion

In order to validate the engineering pipeline and the proposed variable injection strategy, we conducted a micro-scale fine-tuning experiment using 6-hourly ERA5 data from January 2015. Although this extremely limited temporal window is insufficient for establishing climatological skill scores (which require multi-decadal evaluation), it serves as a critical unit test for the model's ability to learn MJO dynamics from raw physical fields and propagate them using the pre-trained backbone.

A. Training Dynamics and Convergence

The training process was stable, which shows that our `OneCycleLR` scheduling and gradient accumulation strategy worked well. Figure 1 illustrates the evolution of training and validation loss over 30 epochs.

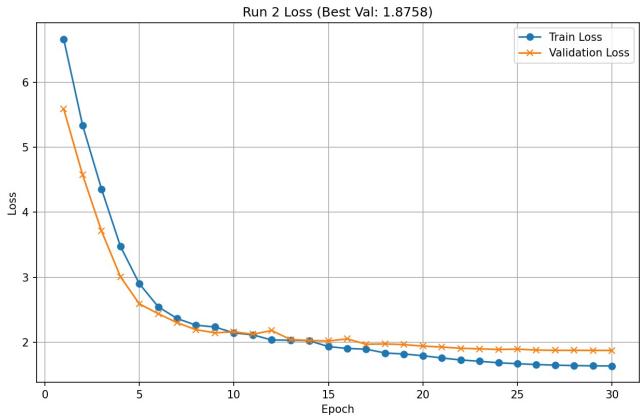

Fig. 1: Training and validation loss trajectories using the `OneCycleLR` scheduler. The smooth convergence of the validation loss (orange) to ≈ 1.87 indicates successful domain adaptation without overfitting, despite the limited dataset size.

In preliminary tests using cosine annealing (loss curve not shown), we observed quite a lot of volatility in the loss landscape, characterized by *zig-zagging* optimization paths. This suggested that the model was struggling to settle into a minimum compatible with both the pre-trained weights and new variable embeddings—i.e., it was not successfully *learning*. By contrast, the run presented here shows a monotonic decrease in error during the initial high-LR phase, followed by stable convergence. The alignment between validation and training losses tells us that the model is learning generalized physics and not simply memorizing noise.

B. Physical Fidelity of Convective Structures

For this proof-of-concept, our main goal was to see if the model could produce realistic convective structures. Standard regression models often minimize error by predicting a smoothed average of all possible future states, but in our case this would result in blurry and ultimately unphysical forecasts.

Figure 2 compares the Ground Truth OLR (top) with the Aurora forecast (middle) at a 6-hour lead time. The model captures the spatial structure of tropical convection well. It retains the high-frequency texture and sharp gradients characteristic of actual cloud systems; it does not degenerate into a Gaussian blur.

The Difference Map (Figure 2, bottom) reveals a specific error topology characterized by paired regions of high positive and negative error (red/blue dipoles). In meteorological verification, this is known as the "double penalty" effect. When a model predicts a sharp physical feature (e.g., a convective cluster) that is slightly displaced in space relative to the ground truth, it is penalized twice: once for missing the feature at the correct location, and once for placing it in the wrong location. This results in a higher RMS error compared to a blurry forecast. The double penalty nonetheless is an indicator of physical realism. It confirms that the Spectral Loss term $\mathcal{L}_{\text{spec}}$ encouraged the model to generate realistic small-scale structures.

C. Autoregressive Rollout Stability

To assess the stability of the fine-tuned dynamical core, we performed a 10-day (40-step) autoregressive rollout initialized from the validation set (January 21st, 2015). It is crucial to note that the model was only trained on single step ($t \to t+6h$) predictions; therefore, its ability to remain stable over 40 steps is a test of its generalization to the underlying physics of the system.

Figure 3 shows the Root MSE (RMSE) of the TTR variable over the 10-day horizon. The error growth is monotonic and broadly linear after $N = 10$ steps (2.5 days), reaching ~ 34 Wm^{-2} at Day 10. The error does

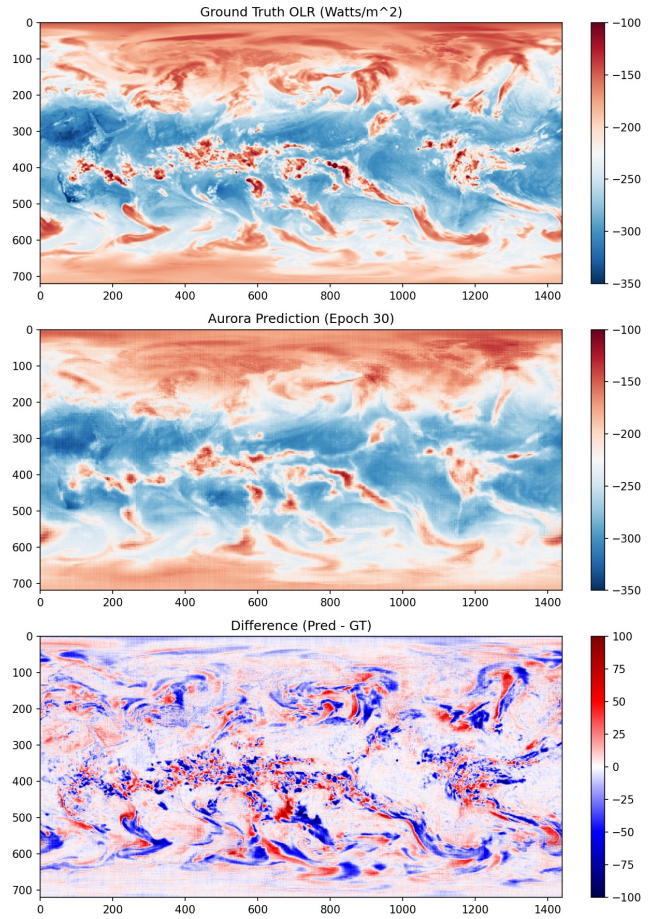

Fig. 2: Visualizing Convection via OLR (Watts/m^2) at Epoch 30. **Top:** Ground Truth ERA5 data. **Middle:** Aurora Forecast. **Bottom:** Difference Map (Prediction - Ground Truth). The model successfully generates sharp convective structures. The difference map reveals dipole error patterns (red/blue pairs), a signature of the "double penalty" effect caused by slight spatial displacements of sharp features.

not explode exponentially, nor does the model collapse to a simple climatological mean.

Figure 4 visualizes the forecast evolution. At Day 1, the forecast is sharp and highly accurate. By Day 5, we observe phase errors (spatial displacement of convective clusters), but the structural integrity of the MJO signal is preserved. By Day 10, although the macroscopic envelope of the MJO is still visible, the field exhibits high-frequency noise or "grittiness." This artifact is likely a consequence of the Spectral Loss weight ($\lambda = 0.05$); while it is effective in enforcing sharpness at $t = 1$, unconstrained autoregression allows high-frequency errors to accumulate over time.

This result validates our proposed two-stage curriculum. Stage 1 has successfully taught the model the *local* physics and variable distributions. The accumulation of noise at Day 10 provides the empirical justification for Stage 2, where optimizing Low-Rank Adapters on multi-step rollouts will specifically target the suppression of these drifting error modes.

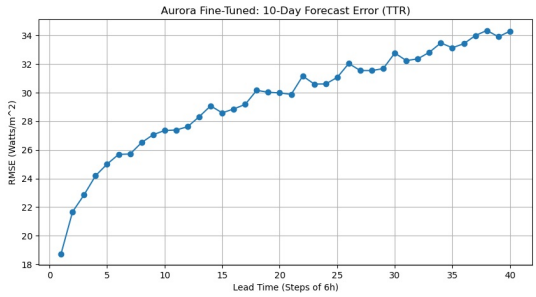

Fig. 3: RMSE of Top Thermal Radiation (TTR) over a 10-day autoregressive forecast. The error growth is bounded and linear, indicating that the dynamical core remains stable even though it was only fine-tuned on single-step objectives.

V. Conclusion

In this project, we showed how a large AI weather model can be adapted to the challenges of sub-seasonal forecasting. By reformulating the MJO prediction task from a statistical emulation problem to a prognostic initial value problem, we demonstrated that the Microsoft Aurora foeundation model can be fine-tuned to ingest and propagate critical thermodynamic variables—viz., Outgoing Longwave Radiation (OLR/TTR) and Total Column Water Vapor (TCWV)—that were absent from its pre-training corpus.

Our micro-scale fine-tuning experiment serves as a robust validation of the proposed engineering pipeline. We showed that a curriculum learning approach, using a `OneCycleLR` scheduler and a spectral consistency loss, allows the model to adapt to new variable distributions without suffering from catastrophic forgetting. Importantly, the model generates physically realistic and sharp convective structures rather than the blurry averages typical of standard regression approaches. The error in our 10-day forecast grew linearly. This confirms the fine-tuned model is stable, though the accumulation of high-frequency noise highlights the specific necessity of multi-step optimization in future work to come.

As this project evolves into Phase 2, our focus shifts from architectural adaptation to climatological generalization. The immediate priority is to scale the training

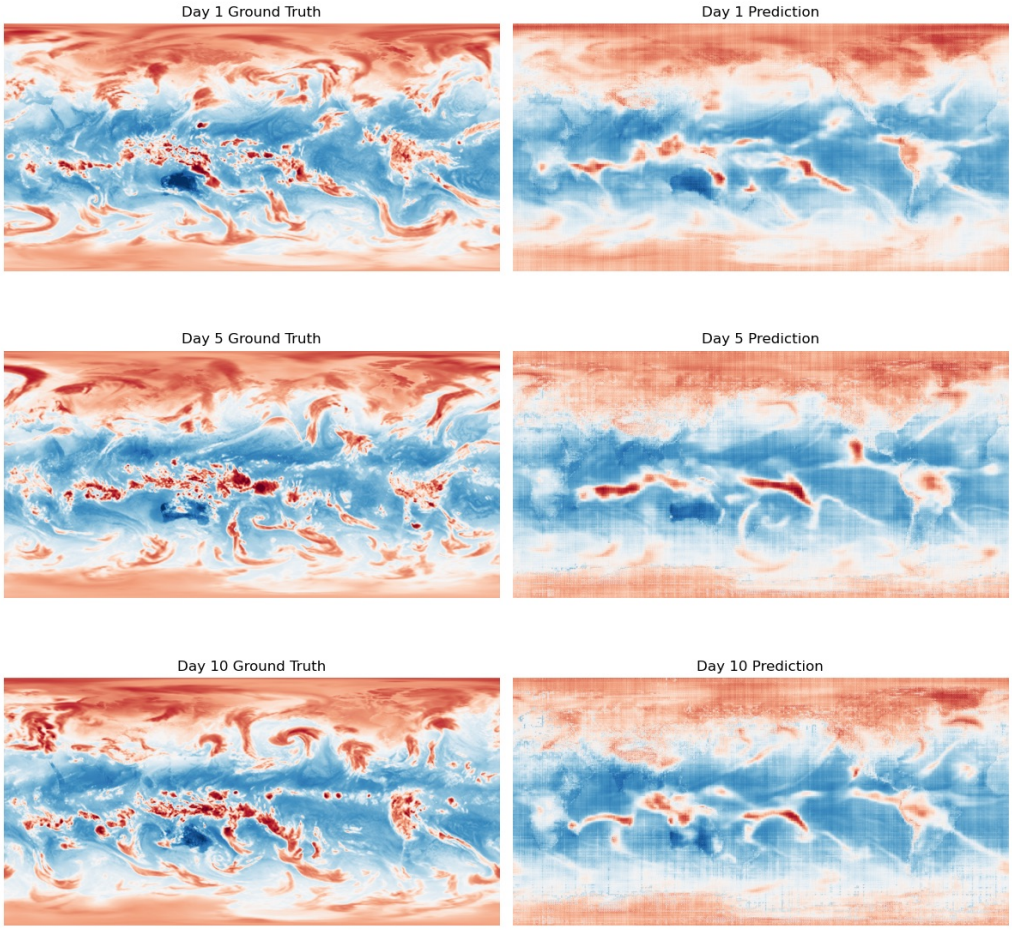

Fig. 4: Evolution of the forecast over 10 days. **Left:** Ground Truth. **Right:** Aurora Prediction. While the model maintains physical realism through Day 5, the accumulation of high-frequency noise by Day 10 (bottom right) highlights the necessity for the proposed Stage 2 training (multi-step LoRA) to constrain long-term rollouts.

dataset from the current one-month window to a fuller, multi-decadal dataset. This will be essential for capturing the complex inter-annual variability of the MJO and its modulation and climate drivers such as ENSO, as well as allowing the possibility to compute the RMM indices.

To address the noise accumulation that we observed in long rollouts (see figure 4), we will implement Stage 2 of our training strategy: Low-Rank Adaptation (LoRA) [10], [11] on autoregressive trajectories. By freezing the 1.3B parameters of the backbone and training only low-rank matrices injected into the attention layers, we can computationally afford to optimize the model over 30-day forecast horizons directly. This will allow the model to learn to suppress error growth and drift over time, again enabling satisfactory quantitative benchmarking against the standard RMM indices.

Finally, leveraging the expertise of the Lu Group in Scientific Machine Learning (SciML), we intend to explore physics-informed constraints to further stabilize these long-range forecasts. This may involve augmenting the loss function with penalties for violating the conservation of mass or energy, or training a lightweight residual corrector network to minimize physical residuals. Ultimately, this work suggests that foundation models offer a viable, data-efficient path to bridging the sub-seasonal predictability gap, combining the generalizability of large-scale pre-training with the specific physical constraints of tropical meteorology.

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